The Kohler–Jobin transfer: sharp Faber–Krahn stability from sharp Saint–Venant stability

Plan 1 / Route A, building block KohlerJobinTransfer

Abstract

We record the Kohler–Jobin transfer step. *Input:* the sharp Saint–Venant stability inequality

$$E(\Omega) - E(B^*) \geq c_{\mathrm{SV}}(N) \left(\frac{|B^*|}{|B_1|}\right)^{(N+2)/N} \mathcal{A}(\Omega)^2 \qquad \text{for every open } \Omega,\ |\Omega| = |B^*|,$$

produced by the BoundedReduction block from the sibling SaintVenantAssembly block. *Output:* the sharp Faber–Krahn stability inequality

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/N} \big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(N)\ \mathcal{A}(\Omega)^2.$$

The argument is non-perturbative: it uses only the Kohler–Jobin inequality $T \cdot \lambda_1^{(N+2)/2} \geq T_N L_N^{(N+2)/2}$ (cited from Brasco [1]) and one elementary one-variable inequality. No compactness, no contradiction, no Taylor expansion.

1 Notation

For open bounded $\Omega \subset \mathbb{R}^N$ $(N \geq 2)$:

- $u_\Omega \in H_0^1(\Omega)$ is the torsion potential, the unique weak solution of $-\Delta u = 1$ in Ω, $u = 0$ on $\partial\Omega$;
- $T(\Omega) = \int_\Omega u_\Omega \, dx$ is the torsional rigidity;
- $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue;
- $E(\Omega) := -T(\Omega)/2$ is the BDV-normalized Saint–Venant energy, so $T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*))$;
- B^* is the ball with $|B^*| = |\Omega|$, B_R is the ball of radius R at the origin;
- $\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^N} |\Omega \Delta (x_0 + B^*)|/|B^*|$ is the Fraenkel asymmetry;
- B_1 is the unit ball, $L_N := \lambda_1(B_1)$, $T_N := T(B_1) = |B_1|/(N(N+2))$.

The scaling laws

$$T(t\Omega) = t^{N+2} T(\Omega), \qquad \lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega) \qquad (t > 0), \tag{1}$$

make $\Psi(\Omega) := T(\Omega) \cdot \lambda_1(\Omega)^{(N+2)/2}$ scale invariant. Write $\beta := (N+2)/2$, $L := \lambda_1(B^*)$, $T_0 := T(B^*)$.

2 The Kohler–Jobin inequality

Theorem 1 (Kohler–Jobin, 1978). *For every open bounded $\Omega \subset \mathbb{R}^N$ with $\lambda_1(\Omega) < \infty$ and every ball B,*

$$T(\Omega) \cdot \lambda_1(\Omega)^\beta \geq T(B) \cdot \lambda_1(B)^\beta. \tag{2}$$

Equivalently $\Psi(\Omega) \geq \Psi(B_1) = T_N L_N^\beta$, with equality iff Ω is a ball (up to translation and a null set).

We invoke (2) as a black box; for a modern proof by Schwarz symmetrization see Brasco [1, Theorem 3.3].

Remark 2 (Scaling check). $\beta = (N+2)/2$ is forced by scale-invariance of (2). From (1), $T(t\Omega)\lambda_1(t\Omega)^\beta = t^{N+2-2\beta} T(\Omega)\lambda_1(\Omega)^\beta$; invariance forces $N + 2 - 2\beta = 0$.

3 Pointwise transfer at fixed volume

Lemma 3 (Pointwise Kohler–Jobin transfer). *For every open $\Omega \subset \mathbb{R}^N$ with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) \geq L \cdot \Big(\frac{T_0}{T(\Omega)}\Big)^{1/\beta} = L \cdot \Big(\frac{T_0}{T(\Omega)}\Big)^{2/(N+2)}. \tag{3}$$

Proof. Apply Theorem 1 with $B = B^*$ and rearrange. □

By the classical Saint–Venant inequality $T(\Omega) \leq T_0$ at fixed volume, the Saint–Venant deficit $\delta_{\mathrm{SV}}(\Omega) := T_0 - T(\Omega) \geq 0$ is non-negative. Set $s := \delta_{\mathrm{SV}}/T_0 \in [0, 1)$; (3) becomes

$$\lambda_1(\Omega) - L \geq L\big[(1-s)^{-1/\beta} - 1\big]. \tag{4}$$

4 An elementary one-variable inequality

Lemma 4. *For $p \in (0, 1]$, $s \in [0, 1)$,*

$$(1-s)^{-p} - 1 \geq p\, s. \tag{5}$$

Proof. $f(s) := (1-s)^{-p} - 1$ satisfies $f(0) = 0$ and $f'(s) = p(1-s)^{-(p+1)} \geq p$ for $s \geq 0$. By FTC, $f(s) = \int_0^s f'(\tau)d\tau \geq ps$. □

Remark 5. No Taylor expansion; the inequality is exact on $[0, 1)$. The factor p is sharp at $s = 0$ since $f'(0) = p$. We use $p = 1/\beta = 2/(N + 2) \in (0, 1]$.

5 Small-deficit linearization

Lemma 6 (Small-deficit linearization). *For every open $\Omega \subset \mathbb{R}^N$ with $|\Omega| = |B^*|$ and $\delta_{\mathrm{SV}}(\Omega) \leq T_0/2$,*

$$\lambda_1(\Omega) - L \geq \frac{2L}{(N+2)T_0}\, \delta_{\mathrm{SV}}(\Omega). \tag{6}$$

Proof. $s = \delta_{\mathrm{SV}}/T_0 \leq 1/2$, $p = 2/(N + 2)$. By (4) and Lemma 4, $\lambda_1(\Omega) - L \geq L[(1-s)^{-p} - 1] \geq Lps = Lp\delta_{\mathrm{SV}}/T_0 = (2L/((N + 2)T_0))\delta_{\mathrm{SV}}$. □

The constant $2L/((N + 2)T_0)$ is the universal *Kohler–Jobin multiplier*. Both sides of (6) scale, so it depends only on N: $2L/((N + 2)T_0) = 2L_N/((N + 2)T_N) = 2NL_N/|B_1|$.

6 Large-deficit branch

Lemma 7 (Large-deficit bound)**.** *If* $|\Omega| = |B^*|$ *and* $\delta_{\mathrm{SV}}(\Omega) > T_0/2$*,*

$$\lambda_1(\Omega) - L \geq L\big(2^{2/(N+2)} - 1\big). \tag{7}$$

Proof. $T_0/T(\Omega) > 2$, so (3) gives $\lambda_1(\Omega) \geq L \cdot 2^{2/(N+2)}$. □

Since $\mathcal{A}(\Omega) \in [0, 2)$ for $|\Omega| = |B^*|$, $\mathcal{A}(\Omega)^2 \leq 4$, so (7) yields in this regime

$$\lambda_1(\Omega) - L \geq \frac{L(2^{2/(N+2)} - 1)}{4}\, \mathcal{A}(\Omega)^2. \tag{8}$$

7 Main transfer theorem

Theorem 8 (Sharp Faber–Krahn via Kohler–Jobin)**.** *Assume sharp Saint–Venant stability in energy form: there exists* $c_{\mathrm{SV}}(N) > 0$ *such that*

$$E(\Omega) - E(B^*) \geq c_{\mathrm{SV}}(N)\Big(\frac{|B^*|}{|B_1|}\Big)^{(N+2)/N} \mathcal{A}(\Omega)^2 \quad \forall\,\Omega,\ 0 < |\Omega| = |B^*| < \infty. \tag{9}$$

Then for every such Ω*,*

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/N} \big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(N)\ \mathcal{A}(\Omega)^2, \tag{10}$$

with

$$c_{\mathrm{FK}}(N) := \min\left\{\frac{4L_N\, c_{\mathrm{SV}}(N)}{(N+2)T_N},\ \ \frac{L_N(2^{2/(N+2)} - 1)}{4}\right\}. \tag{11}$$

Proof. Split on $\delta_{\mathrm{SV}} \leq T_0/2$ vs. $\delta_{\mathrm{SV}} > T_0/2$.

Case 1. Put $r = (|B^*|/|B_1|)^{1/N}$. Lemma 6 and (9) give $\lambda_1(\Omega) - L \geq (2L/((N+2)T_0))\delta_{\mathrm{SV}} \geq (4L\, c_{\mathrm{SV}} r^{N+2}/((N+2)T_0))\,\mathcal{A}^2 = (4L_N c_{\mathrm{SV}}/((N+2)T_N))r^{-2}\,\mathcal{A}^2$ by scaling. Multiplying by $r^2 = (|\Omega|/|B_1|)^{2/N}$ gives the first constant in (11).

Case 2. (8) gives $\lambda_1(\Omega) - L \geq r^{-2}(L_N(2^{2/(N+2)} - 1)/4)\,\mathcal{A}^2$ by scaling. Multiplying by r^2 gives the second constant.

Either way, the conclusion holds with (11). □

Remark 9 (Energy normalization)**.** With $E := -T/2$, the torsion deficit is $\delta_{\mathrm{SV}} = T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*))$. This is the source of the factor $4L_N/((N+2)T_N)$ in the small-deficit branch of (11).

Remark 10 (Sharpness)**.** The exponent 2 on $\mathcal{A}$ in (10) is sharp; it is saturated by smooth ellipsoidal perturbations of the ball where $\lambda_1(\Omega) - L$ and $\mathcal{A}^2$ are of the same order ([1] and references).

8 Summary of constants

Symbol	Value / dependence	Source
N	dimension, ≥ 2	hypothesis
L_N	$\lambda_1(B_1) = j^2_{N/2-1,1}$	
T_N	$T(B_1) = \lvert B_1\rvert/(N(N+2))$	
$\beta = (N+2)/2$	scaling exponent	Rmk. 2
L, T_0	$\lambda_1(B^*), T(B^*)$	
$2L_N/((N+2)T_N) = 2NL_N/\lvert B_1\rvert$	KJ multiplier	Lem. 6
$L_N(2^{2/(N+2)} - 1)$	large-deficit gap	Lem. 7
$c_{\mathrm{SV}}(N)$	global sharp Saint–Venant energy constant	BoundedReduction
$c_1(N)$	$4L_N c_{\mathrm{SV}}/((N+2)T_N)$	small branch
$c_2(N)$	$L_N(2^{2/(N+2)} - 1)/4$	large branch
$c_{\mathrm{FK}}(N)$	$\min\{c_1(N), c_2(N)\}$	Thm. 8

All constants are explicit; none depends on a compactness or contradiction step in the Kohler–Jobin transfer.

References

[1] L. Brasco, *On torsional rigidity and principal frequencies: an invitation to the Kohler–Jobin rearrangement technique*, ESAIM Control Optim. Calc. Var. **20** (2014), 315–338.